Private Exposure Registry · One-page spec

Case 02 · A Slow-Motion Crisis · Stage 04: Specification

The problem

The biggest danger is the exposure nobody saw. Each regulator sees its own institutions, and no fund will reveal its positions to competitors. The result is a market where everyone knows everything about themselves and nothing about the whole.

The promise

One picture of the whole market, built from data nobody sees; built on a standard already running in the world, not on cryptography we invented; and every number in it can be checked, not just trusted.

Ten parts in four layers

- **At the participant:** LEI-based onboarding with one blind token per institution (1). Local encoding of the loan book into a fixed table (2). A commitment per cell, and a signed receipt (3).
- **Between the servers:** three-way sharding with validity proofs, per Prio3 (4). Aggregation across three servers at three regulators (5). Every sum checked against the commitments (6).
- **The gate:** a closed query catalog, a threshold of 5 participants per cell, and a privacy budget (7). Encrypted envelopes per server, and operations that prefer delay to leakage (8).
- **Transparency:** an append-only audit log with random audits (9). An exact dashboard for regulators, and a public report with error margins (10).

Six decisions

- **All three servers are required:** a delayed report beats a leaked secret.
- An IETF standard and production code, not our own protocol.
- Only supervised institutions report, with tokens servers cannot link.
- Verify the servers instead of trusting them.
- A closed query catalog instead of free-form questions.
- Exact numbers for regulators; noisy numbers with error margins for the public.

Still open

- **A lying participant:** proofs show range, not truth. The system is only as reliable as the regulatory reporting beneath it.
- The system shows today's picture, not what will happen to it.